



Francisco Javier Guinot Almenar

Software Engineer | Game Engine Developer | Low-Level Programmer

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Skills

📁 Programming Languages

Advanced
C, C++, C#, Python, SQL, GLSL, Bash / Shell Scripting, Batch Scripting

🔧 Graphics & Low-Level

Advanced
Vulkan, OpenGL, Memory Management, Multi-threading, Engine Development, Real-time Simulation

📦 Tools & Frameworks

Advanced
Git, CMake, Visual Studio, Unity, Unreal Engine 5, Linux, Jira, Subversion (SVN), Premake

Projects

Vulkan Graphics Engine (Ongoing)

- Developing a custom graphics engine from the ground up using the Vulkan API.
- Manual memory allocation, synchronization primitives, and graphics pipelines to achieve maximum hardware control.
- Cross-Platform: Available for PC (Windows / Linux / MacOS) and Mobile (Android)

OpenGL Custom Framework Sep 2023 - Jun 2024

- Created a 3D rendering framework in C++ without third-party engines.
- Built custom math libraries (Linear Algebra) and shader management systems (GLSL).
- [GitHub project \(Public\)](#)

Languages

Spanish
Native

English
Professional Working Proficiency

Summary

Performance-oriented Software Engineer with a background in **high-fidelity simulation** and **game engine development**. Specialized in C, C++, and C#, with a strong focus on writing "raw" code, minimizing external dependencies, and optimizing for hardware. Experienced in real-time graphics (OpenGL/Vulkan) and developing complex backend systems for professional training simulators.

Technical Philosophy

I advocate for minimal-dependency development and hardware-level optimization while maintaining a strict focus on code safety and stability. I prioritize writing robust, deterministic software that prevents common pitfalls such as memory corruption, pointer mismanagement, or unhandled exceptions. My goal is to build systems that are both ultra-efficient and architecturally sound, ensuring that high performance never comes at the cost of reliability.

Education

ESAT (Escuela Superior de Arte y Tecnología) Game Programming	HND in Computing and Systems Development Valencia • Sep 2021 - Jun 2024
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- Intensive focus on **Advanced C++**, Data Structures, Computer Architecture and Gaming engines (Unity / Unreal Engine 5)
- Developed custom game engines from scratch using C++ and Graphics APIs (OpenGL/Vulkan).
- Gained a deep understanding of memory management, optimization, and software engineering patterns applied to real-time systems.

E.S.A.T Web

Experience

Simmodar Training Solutions (LSyM - University of Valencia) Software Developer / Systems Programmer	Burjassot, Valencia Oct 2024 - Present
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- Developed and maintained high-performance **real-time simulation** software for professional training.
- Implemented core logic and backend systems using **C++ and C#**, focusing on low-latency response and physical accuracy.
- Worked on the integration of hardware-in-the-loop systems and optimized rendering pipelines to ensure stable frame rates in complex simulation environments.

LSyM Web

Interests

🔧 **Programming**
Low-level Hardware Architecture, Retro-computing & Emulation, Mathematics for Computer Graphics, Physics Simulations

Personal webs

Linkedin

Github
Portfolio: gitjvguinot.github.io